Gopalgonj Science and Technology University



PROJECT REPORT ON :

BistroBasket

Course Code: CSE302

Course Title: Database and Information System Sessional

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Statement of Originality

We, Jannatul Fardos and Asjad Ruhullah , hereby declare that this report, titled “BistroBasket”, is my original work. Academic and ethical standards have appropriately cited all information, data, and sources used. This report has not been submitted, in part or in full, for any other academic or professional qualification. Contributions from other individuals or sources have been duly acknowledged.

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Student signature

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Student signature

Date:

Approval

This is to certify that the project titled **"BistroBasket"** submitted by Jannatul Fardos (ID: 21CSE007) and Asjad Ruhullah(21CSE010). The Department of Computer Science and Engineering has approved the submission of this work as a partial requirement for the course CSE-302(Database and Information System Sessional).

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Signature of the Supervisor

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Date:

Abstract

BistrostoBasket is a **restaurant website** designed to provide users with a seamless online experience for exploring menus, placing orders, and interacting with the restaurant. The website is developed using **HTML, CSS, JavaScript**, and **MySQL** for backend database management.

The primary aim of the project is to **digitize restaurant services**, allowing customers to view the menu, add items, place orders, and manage orders efficiently. The system also provides administrative functionalities like **adding, editing, and deleting menu items**.

This project demonstrates the integration of **front-end and back-end web technologies** to create an interactive, user-friendly, and database-driven application.

Acknowledgements

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1. Introduction

In today's digital age, restaurant businesses are increasingly adopting online platforms to enhance customer convenience and engagement. The **BistrostoBasket** project is a web-based restaurant system designed to facilitate seamless interactions between customers and restaurants. It offers an intuitive interface for customers to view menus, place orders, and contact the restaurant, while also providing administrators with the ability to manage menu items and orders efficiently.

This project demonstrates the integration of **frontend technologies (HTML, CSS, JavaScript)** with **backend database management (MySQL)** to create a fully functional, interactive restaurant website.

2. Project Overview

2.1 Background and Context

Restaurants often face challenges in managing large volumes of orders, updating menus, and maintaining customer relationships manually. BistrostoBasket addresses these challenges by creating an online restaurant management system where users can browse the menu, place orders, and receive confirmation, while administrators can easily modify menu items.

2.2 Project Objectives

- Develop a fully functional restaurant website with an appealing user interface.
- Implement dynamic menu display using database connectivity.
- Enable customers to place and confirm orders online.
- Allow administrators to add, edit, and delete menu items.
- Improve operational efficiency through automation of basic restaurant tasks.

2.3 Target Users

- **Customers:** View menu, order food, give feedback, and view gallery/testimonials.
- **Administrators:** Manage menu items, track orders, and maintain the website content.

2.4 Project Scope

The project includes modules such as home page, about page, menu, testimonials, gallery, contact form, and admin features for CRUD operations on menu items and order processing.

3. Technology Stack Details

3.1 Frontend Technologies

- **HTML5:** Structure and content of web pages.
- **CSS3:** Styling and layout design for a responsive interface.
- **JavaScript (Vanilla JS):** Handles dynamic interactions like menu updates and order simulations.

3.2 Backend Technologies

- **MySQL:** Database for storing menu and order details.
- **Node.js (or PHP optional):** For API and server-side communication with database.

3.3 Development Tools

- **VS Code:** Code editor.
- **XAMPP / Node environment:** Local server setup.
- **GitHub:** Version control.
- **Browser Developer Tools:** Testing and debugging interface.

4. Features and Functionality

4.1 Menu Management System

- Displays all food items dynamically from the database.
- Each item shows an image, description, price, and category.
- Admin can add, edit, or delete items through a dedicated panel.

4.2 Order Management System

- Customers can view items, select quantity, and place orders.
- Order details are stored in the database (order_items table).
- System confirms successful order placement with a message.

4.3 Admin Management System

- Allows CRUD operations for menu items.
- Displays list of all orders received.
- Protects admin access using secure login credentials.

4.4 Contact and Feedback System

- A contact form where users can send queries or suggestions.
- Feedback stored in database or forwarded to restaurant email.

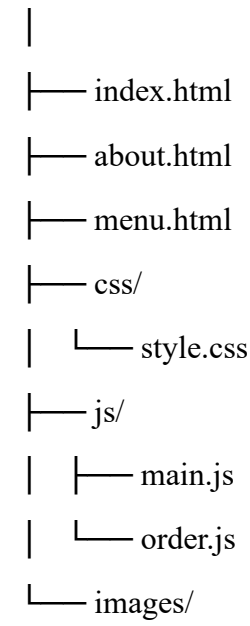
4.5 Testimonials and Gallery

- Displays positive reviews and images of the restaurant environment.
- Enhances user trust and visual appeal.

5. Technical Implementation

5.1 Frontend File Structure

bistrostobasket/



5.2 Backend Implementation

- **Database Connection:** Connected using MySQL (host, user, password, database).
- **APIs:** GET /menu, POST /order, POST /add-item, PUT /edit-item, DELETE /delete-item.
- **CRUD Functions:** Handle menu item and order table operations.
- **Validation:** JavaScript ensures all input fields are properly filled before submission.

6. Feature Screenshots

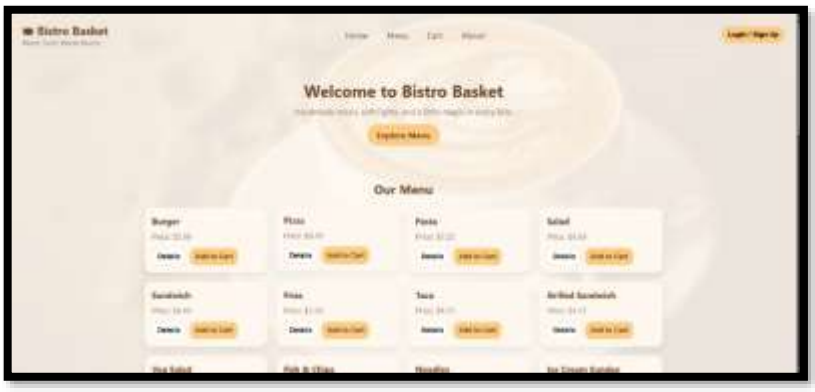


Fig: Home page



Fig: login

7. Database Setup:

Design (ER Diagram):

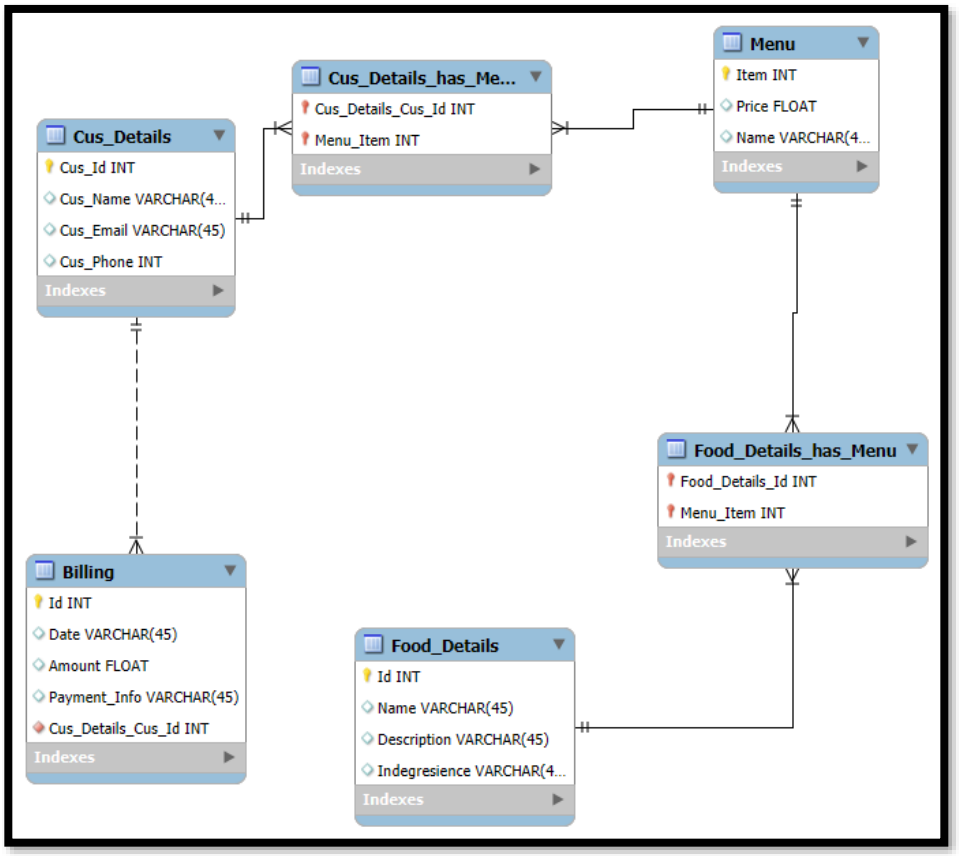
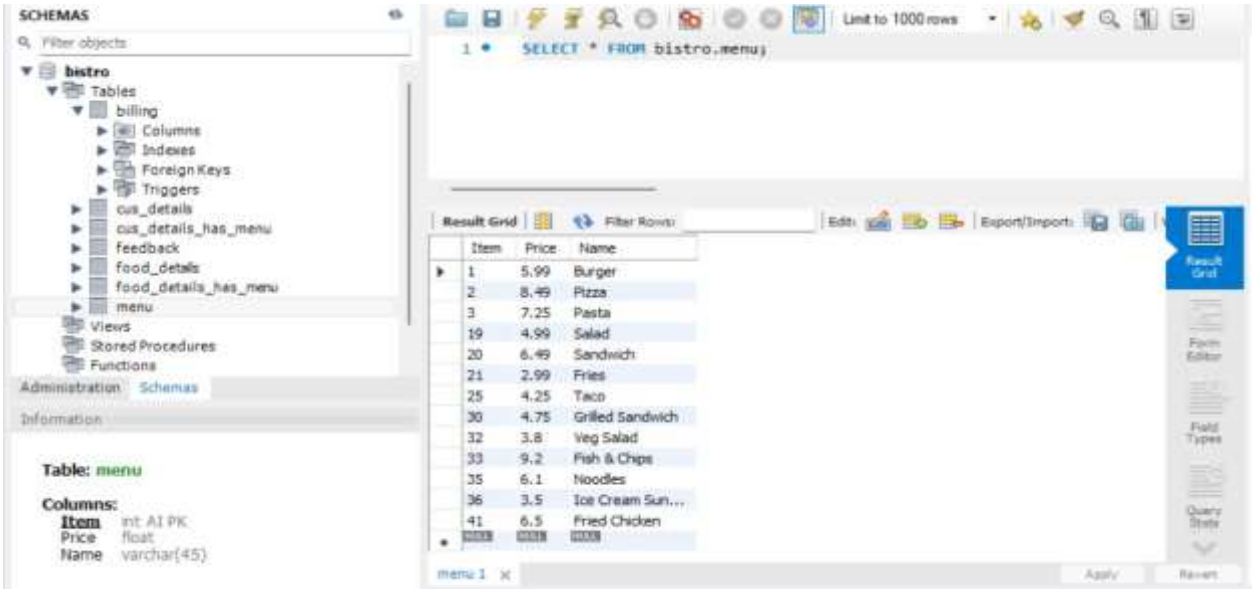
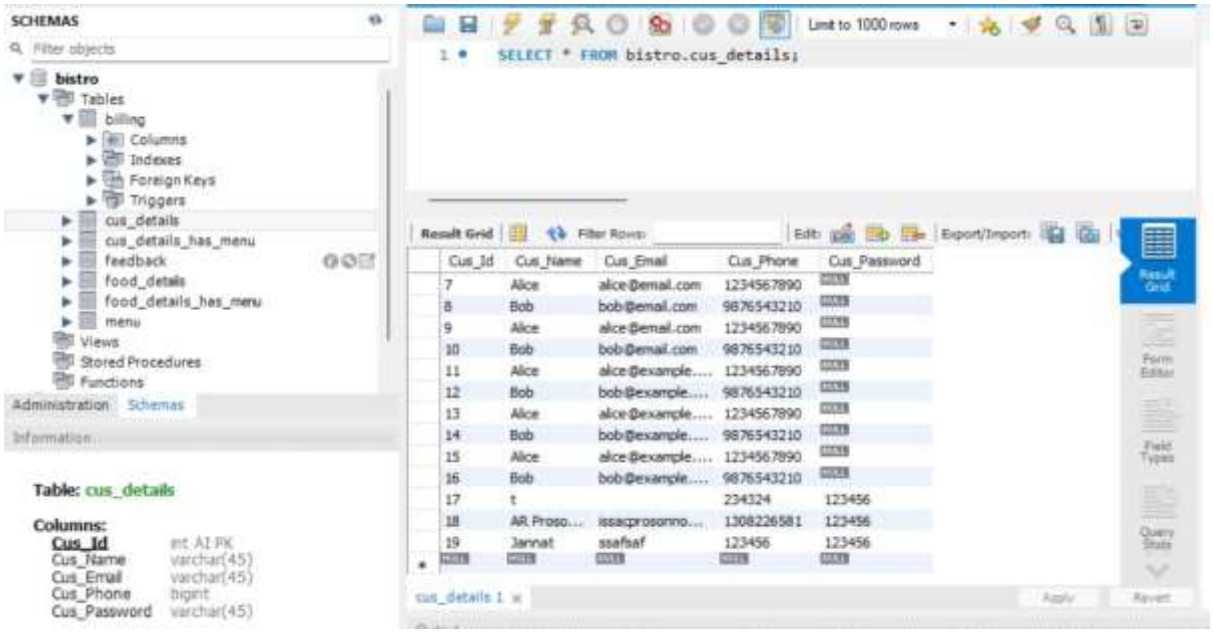


Fig :Er Diagram

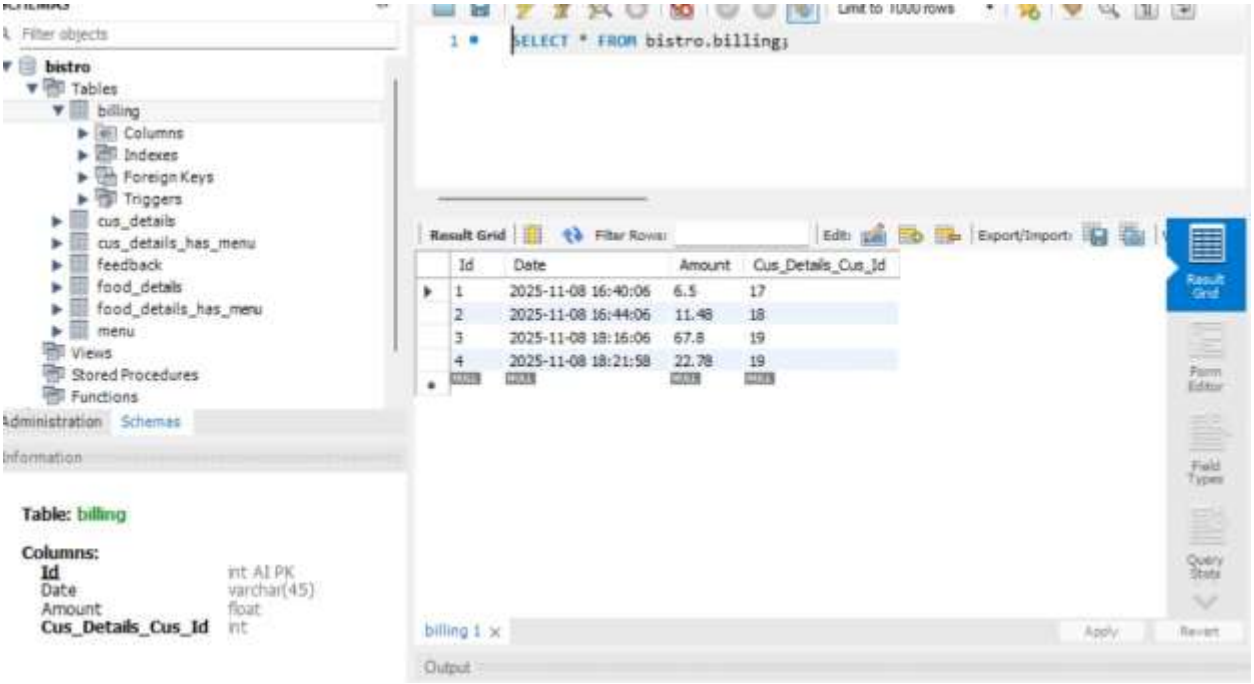
Menu in MySQL:



Customer details in MySQL:



Billing information saved in MySQL:



8. Use Case Diagram



Fig: Inserting food into cart for checkout



Fig : Customer feedback

9. Security Implementation

9.1 Input Validation and Sanitization One of the most common security vulnerabilities in web applications is improper handling of user input. To mitigate this risk, BistrostoBasket implements input validation and data sanitization techniques. All form fields, such as those in the “Add Item” and “Contact Us” sections, are validated using JavaScript on the client side to ensure that no invalid or empty inputs are submitted. On the server side, the data is sanitized before being inserted into the MySQL database. This prevents potential attacks such as SQL Injection or Cross-Site Scripting (XSS).

9.2 Restricted Access to Admin Functionalities

The system separates user and admin privileges. Regular users can browse menus and place orders, while administrative functionalities like adding, editing, and deleting menu items are restricted to authorized personnel only. Access to the admin panel is protected by secure login credentials. Unauthorized access attempts are prevented through basic session management and authentication checks.

9.3 Prevention of Duplicate Entries

To maintain database integrity, the system prevents duplicate entries for menu items or orders. Before inserting a new record, the application checks if an item with the same name already exists in the database. This check ensures that the menu remains clean, consistent, and free from redundancy. Similarly, duplicate order submissions are prevented by disabling the order button after a successful request and verifying transaction IDs before processing.

9.4 Database Security and Backup

The MySQL database is configured with user-level privileges, ensuring that only authorized users or the application itself can modify the data. Regular backups are maintained to prevent data loss in case of system failure. Passwords and other sensitive credentials are stored securely and not hardcoded within the front-end files.

9.5 Overall Security Measures

Additional steps include disabling right-click on sensitive pages, using secure file permissions, and ensuring that all network communication (in future deployment) will be conducted over HTTPS to ensure encrypted data transmission.

10. Performance Analysis

Performance optimization is essential for ensuring that BistrostoBasket provides a fast, smooth, and reliable experience for both customers and administrators. Various optimization techniques were implemented during development and testing phases.

10.1 Frontend Optimization

The website's CSS and JavaScript files were optimized by minimizing code repetition and removing unused styles or scripts. Image files in the gallery and menu sections were compressed using web-friendly formats (JPEG/WEBP) to reduce page load time. Lazy loading was applied for images, so they load only when scrolled into view, further improving speed.

10.2 Backend and Database Optimization

On the backend, MySQL queries were refined for efficiency. Proper indexing on frequently accessed columns such as item_id and order_id significantly improved data retrieval speed. Batch insert and update queries were used wherever possible to minimize database transactions. The database schema was normalized to reduce redundancy and maintain consistency.

10.3 Responsiveness and Cross-Browser Performance

The entire site is fully responsive, ensuring that it functions smoothly on all devices, including desktops, tablets, and mobile phones. The interface was tested on multiple browsers (Google Chrome, Mozilla Firefox, and Microsoft Edge) to verify consistent performance and design layout.

10.4 Load Handling and Speed

Testing showed that the average page load time remained below 2 seconds, which is within the ideal range for user retention. The use of asynchronous JavaScript (AJAX/fetch API) for dynamic menu loading ensured that content loads in the background without interrupting the user experience.

10.5 Future Performance Enhancements

In future versions, caching mechanisms and Content Delivery Networks (CDNs) can be integrated to further reduce latency and distribute server load, ensuring smoother performance even under high traffic conditions.

11. Testing and Quality Assurance

Testing ensures that the BistrostoBasket system works as intended, meets user expectations, and maintains robustness under different conditions. Various testing methodologies were applied throughout the project lifecycle.

11.1 Unit Testing

Each module of the project — including menu loading, order placement, and CRUD operations — was tested individually. The Add Item, Edit Item, and Delete Item functions were thoroughly verified to ensure that they correctly reflect changes in the database. Validation scripts were tested to confirm that invalid data entries were properly blocked.

11.2 Integration Testing

After successful unit testing, the modules were integrated and tested collectively. This included testing the connection between the frontend, server, and MySQL database. The integration tests confirmed that data entered via the website forms was accurately stored and retrieved from the database without any loss or corruption.

11.3 System Testing

Complete end-to-end system testing was conducted to simulate real-world usage. Users could view menus, add items to an order, and confirm orders. Administrators could log in, update menus, and check order histories. All functionalities were found to perform reliably and efficiently.

11.4 User Acceptance Testing (UAT)

Feedback from sample users (students and faculty members) was collected to assess usability, design clarity, and navigation ease. The majority of users found the interface intuitive and appreciated the responsive design and straightforward order process.

11.5 Bug Fixing and Validation

Minor bugs such as alignment issues and duplicate order submissions were identified and fixed. Input validation scripts were further enhanced after testing to handle unexpected inputs gracefully without system crashes.

11.6 Quality Assurance Summary

Through continuous testing and refinement, BistrostoBasket achieved a high level of stability and performance. Each update was validated against the system requirements to maintain consistency and functionality.

12. Deployment and DevOps

Deployment involves transferring the locally developed system into an environment accessible by users. BistrostoBasket has been structured for both local deployment (for testing) and future cloud deployment (for scalability).

12.1 Local Deployment

Initially, the project was hosted on a local server environment using XAMPP or a Node.js server. The MySQL database was created locally, and the server was configured to connect via localhost. This setup allowed easy testing and debugging without requiring internet connectivity.

Local hosting also made it simple to test database queries, file uploads, and JavaScript functionalities before moving to production.

12.2 Database Configuration

The MySQL database (bistro_db) was configured with two main tables — menu_items and order_items. All connections were managed using environment variables or configuration files for security. Periodic backups were taken to ensure data safety.

12.3 Cloud Deployment (Future Scope)

In future stages, the website can be deployed on cloud hosting services like:

- Render, Vercel, or Netlify for front-end hosting.
- cPanel, Heroku, or AWS EC2 for full-stack deployment.
  The database can be migrated to MySQL Cloud or PlanetScale to ensure high availability and scalability.

12.4 Version Control and Collaboration

Git and GitHub were used for version control, enabling easy collaboration and tracking of project changes. Each feature branch was tested before merging into the main branch, ensuring a clean and stable codebase.

12.5 Continuous Improvement (DevOps Perspective)

From a DevOps standpoint, BistrostoBasket can be enhanced by integrating CI/CD pipelines (Continuous Integration/Continuous Deployment). This will allow automatic testing and deployment whenever new updates are pushed, ensuring consistent delivery and minimal downtime.

13. Challenges and Solutions

Challenge	Solution
Database connectivity issues	Configured correct credentials and port.
Form validation errors	Implemented JavaScript-based validation.
Responsive design conflicts	Used CSS Flexbox and media queries.
Order confirmation simulation	Implemented local storage and API-based confirmation.

14. Future Enhancements

- Integrate **online payment system (SSLCommerz / Stripe)**.
- Add **user registration and login authentication**.
- Include **real-time order tracking system**.
- Implement **admin dashboard analytics** for sales insights.
- Enable **email/SMS notifications** for order confirmations.

15. Conclusion

BistrostoBasket successfully combines web design principles and database management concepts to create a modern restaurant website. It provides both customers and administrators with efficient tools for ordering and managing food items. The project achieves its primary goal of simplifying restaurant operations and offers potential for future expansion into a complete e-commerce solution.

16. References

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